

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `~/Desktop/hw2-brune-Fall25`
Installed JpegTurbo_jll          v3.1.3+0
Installed InvertedIndices        v1.3.1
Installed Libmount_jll          v2.41.1+0
Installed InlineStrings          v1.4.5
Installed Accessors              v0.1.42
Installed Roots                  v2.2.10
Installed Preferences            v1.5.0
Installed DataFrames             v1.8.0
Installed Fontconfig_jll        v2.17.1+0
Installed SentinelArrays        v1.4.8
Installed WorkerUtilities        v1.6.1
Installed WeakRefStrings        v1.4.2
Installed StatsBase              v0.34.6
Installed Expat_jll             v2.7.1+0
Installed DataStructures        v0.19.1
Installed PooledArrays          v1.4.3
Installed TableTraits           v1.0.1
Installed Plots                  v1.40.19
Installed PrettyTables          v3.0.8
Installed ConstructionBase       v1.6.0
Installed DataValueInterfaces   v1.0.0
Installed CompositionsBase      v0.1.2
Installed InverseFunctions      v0.1.17
Installed Libuuid_jll           v2.41.1+0
Installed Crayons                v4.1.1
Installed OpenSSL_jll           v3.5.2+0
Installed Tables                 v1.12.1
Installed Ghostscript_jll       v9.55.1+0
Installed CommonSolve           v0.2.4
Installed IteratorInterfaceExtensions v1.0.0
Installed Glib_jll              v2.86.0+0
Installed FilePathsBase         v0.9.24
Installed CSV                    v0.10.15
Installed SortingAlgorithms     v1.2.2
Installed Latexify               v0.16.10
Installed StringManipulation    v0.4.1
Precompiling project...
 395.9 ms   WorkerUtilities
 397.7 ms   CommonSolve
 452.8 ms   IteratorInterfaceExtensions
 397.1 ms   DataValueInterfaces
 398.9 ms   InverseFunctions

```

259.8 ms	CompositionsBase
292.6 ms	InvertedIndices
326.3 ms	ConstructionBase
500.0 ms	InlineStrings
399.7 ms	PooledArrays
726.6 ms	Crayons
493.0 ms	Preferences
882.0 ms	SentinelArrays
509.1 ms	TableTraits
683.0 ms	FilePathsBase
344.1 ms	InverseFunctions → InverseFunctionsDatesExt
317.8 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
582.4 ms	InverseFunctions → InverseFunctionsTestExt
1329.6 ms	DataStructures
543.4 ms	Unitful → InverseFunctionsUnitfulExt
383.1 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
395.1 ms	Unitful → ConstructionBaseUnitfulExt
533.3 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
632.1 ms	JLLWrappers
332.9 ms	PrecompileTools
523.2 ms	FilePathsBase → FilePathsBaseMmapExt
564.4 ms	SortingAlgorithms
920.4 ms	Tables
815.5 ms	FilePathsBase → FilePathsBaseTestExt
569.2 ms	OpenSSL_jll
358.5 ms	Graphite2_jll
349.6 ms	Libmount_jll
337.1 ms	EpollShim_jll
384.3 ms	LLVMOpenMP_jll
368.4 ms	Bzip2_jll
347.9 ms	Xorg_libICE_jll
350.9 ms	Xorg_libXau_jll
408.1 ms	libpng_jll
382.6 ms	libfdk_aac_jll
390.3 ms	LERC_jll
426.5 ms	LAME_jll
1917.7 ms	Accessors
394.8 ms	fzf_jll
471.3 ms	JpegTurbo_jll
384.6 ms	Ogg_jll
437.4 ms	XZ_jll
356.8 ms	mtdev_jll
342.8 ms	Xorg_libXdmcp_jll

384.9 ms	x265_jll
396.9 ms	x264_jll
398.1 ms	libaom_jll
417.7 ms	Zstd_jll
384.0 ms	Expat_jll
376.9 ms	LZO_jll
362.7 ms	Opus_jll
365.1 ms	Xorg_xtrans_jll
350.0 ms	libevdev_jll
401.3 ms	Libiconv_jll
355.2 ms	eudev_jll
370.6 ms	Libffi_jll
349.4 ms	Libuuid_jll
386.9 ms	FriBidi_jll
1156.7 ms	RecipesBase
1240.1 ms	StringManipulation
587.4 ms	Pixman_jll
1715.2 ms	StatsBase
330.7 ms	Xorg_libSM_jll
539.6 ms	FreeType2_jll
1719.7 ms	OpenSSL
555.8 ms	Accessors → LinearAlgebraExt
459.9 ms	Accessors → TestExt
2988.2 ms	ColorSchemes
462.1 ms	Accessors → UnitfulExt
693.9 ms	JLFzf
323.2 ms	Xorg_libxcb_jll
640.1 ms	libvorbis_jll
358.4 ms	DBus_jll
1248.2 ms	Ghostscript_jll
344.8 ms	libinput_jll
680.1 ms	GettextRuntime_jll
883.3 ms	Libtiff_jll
341.3 ms	Wayland_jll
2044.5 ms	Fontconfig_jll
2524.1 ms	Roots
357.6 ms	Xorg_xcb_util_jll
416.3 ms	Xorg_libX11_jll
7811.6 ms	Parsers
1105.6 ms	Glib_jll
624.5 ms	Roots → RootsUnitfulExt
357.5 ms	Xorg_xcb_util_image_jll
2546.0 ms	Latexify

363.9 ms	Xorg_xcb_util_keysyms_jll
382.1 ms	Xorg_xcb_util_renderutil_jll
417.7 ms	Xorg_xcb_util_wm_jll
366.2 ms	Xorg_libXrender_jll
344.1 ms	Xorg_libXext_jll
391.1 ms	Xorg_libXfixes_jll
378.0 ms	Xorg_libxkbfile_jll
753.8 ms	InlineStrings → ParsersExt
1155.1 ms	JSON
619.5 ms	Latexify → SparseArraysExt
1098.7 ms	UnitfulLatexify
7031.1 ms	PlotUtils
396.1 ms	Xorg_xcb_util_cursor_jll
381.6 ms	Xorg_libXinerama_jll
393.9 ms	Xorg_libXrandr_jll
397.1 ms	Libglvnd_jll
426.1 ms	Xorg_libXcursor_jll
361.1 ms	Xorg_libXi_jll
385.6 ms	Xorg_xkbcomp_jll
750.8 ms	WeakRefStrings
2923.8 ms	Cairo_jll
2212.8 ms	PlotThemes
12725.1 ms	HTTP
529.7 ms	Xorg_xkeyboard_config_jll
375.6 ms	xkbcommon_jll
420.3 ms	Vulkan_Loader_jll
1045.0 ms	HarfBuzz_jll
3430.2 ms	RecipesPipeline
916.1 ms	libass_jll
612.0 ms	Pango_jll
367.4 ms	libdecor_jll
472.7 ms	GLFW_jll
2728.9 ms	Qt6Base_jll
2124.7 ms	FFMPEG_jll
806.2 ms	Qt6ShaderTools_jll
896.3 ms	FFMPEG
1361.2 ms	GR_jll
1574.3 ms	Qt6Declarative_jll
297.7 ms	Qt6Wayland_jll
2725.9 ms	GR
22841.9 ms	PrettyTables
10462.1 ms	CSV
26778.5 ms	DataFrames

```
1805.9 ms    Latexify → DataFramesExt
36290.7 ms   Plots
3029.8 ms    Plots → UnitfulExt
137 dependencies successfully precompiled in 72 seconds. 82 already precompiled.
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

PROBLEM 1 DIAGRAM:

System Diagram of atmosphere temperature and ocean

PROBLEM 1 ANSWER: The atmospheric CO₂ has a positive effect on global mean temperature because higher concentrations strengthen the greenhouse effects. Temperature has a negative effect on ocean CO₂ concentration through Henry's law, since warmer water holds less dissolved CO₂, reducing uptake or increasing outgassing. Atmospheric CO₂ has a positive effect on ocean CO₂ because higher air concentrations create greater dissolution into the ocean. Ocean CO₂ also has a positive effect on atmospheric CO₂, since higher concentrations can lead to outgassing that increases atmospheric levels.

Together, these relationships form a reinforcing cycle. Rising atmospheric CO₂ raises temperature, which decreases the ocean's ability to absorb CO₂, causing more CO₂ to remain in or return to the atmosphere, which further raises temperature. This shows a positive feedback between temperature and the ocean carbon cycle where warming diminishes the ocean's role as a carbon sink and instead promotes additional atmospheric CO₂ accumulation, increasing climate change.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

PROBLEM 2.1 DIAGRAM:

PROBLEM 2.1 ANSWER: I divided the river into three control volumes because each box should begin or end where there is a discontinuity in flow or contaminant input. The first control volume extends from the upstream river inflow to Source A, since that is the first point where a new discharge changes the system. The second volume spans the 15 km between Source A and Source B, where only advection, decay, and deposition occur. At Source B, another discrete discharge enters, creating a new discontinuity and defining the start of the third control volume, which represents the downstream section where the combined plume continues to decay and deposit.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg}/(1000\text{m}^3)$. You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/\text{°C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

References

List any external references consulted, including classmates.